

Retail Sales Analysis Case Study

Project Overview

This case study presents **Retail Sales Analysis** project developed using **Power BI, Excel, and SQL**. The project focuses on analyzing retail sales performance, customer purchasing behavior, product category performance, and profitability to generate business insights through interactive dashboards.

The analysis is based on a retail sales dataset containing **2,000 customer orders**, covering product sales, revenue, profit, customer details, payment methods, discounts, ratings, and monthly sales trends.

Analysis Objective

The objective of this project is to analyze retail sales data and develop an interactive dashboard that helps stakeholders monitor business performance and identify growth opportunities.

- Analyze sales and profit across product categories.
 - Identify top-performing cities based on revenue.
 - Monitor monthly sales trends.
 - Evaluate customer purchasing behavior.
 - Track key business KPIs through interactive dashboards.
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Dataset Description

- **Total Records:** 2,000 Orders
- **Data Source:** Retail Sales Dataset
- **Tools Used:** Excel, SQL, Power BI

Key Attributes

- Order_ID

- Order_Date
- Customer_ID
- City
- Category
- Product_Name
- Quantity
- Unit_Price
- Revenue
- Cost
- Profit
- Payment_Method
- Sales_Channel
- Customer_Rating
- Customer_Type
- Discount_Percentage
- Month

The Retail Sales dataset contains transactional information related to customer purchases, product sales, pricing, profitability, and customer behavior. The data was cleaned and prepared before analysis to ensure accurate reporting and dashboard creation.

Tools & Technologies Used

- **Database:** MySQL – Data storage and SQL analysis
 - **SQL:** Joins, Aggregate Functions, Window Functions, CTE
 - **Data Handling:** CSV Files
 - **Google Sheets:** Formulas, Pivot Tables, Charts
 - **Power BI:** Interactive dashboards and KPI reporting
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Analysis Performed

The retail sales dataset was analyzed using **SQL, Excel, and Power BI** to evaluate sales performance, customer behavior, product profitability, and business growth through data-driven insights.

1. Sales Analysis

- Analyzed revenue and profit across different product categories.
- Identified the highest revenue-generating categories and products.

2. Customer Analysis

- Analyzed customer types and purchasing behavior.
- Compared sales performance between new and returning customers.

3. Regional Analysis

- Evaluated sales performance across different cities.
- Identified top-performing and low-performing locations based on revenue.

4. Product & Profitability Analysis

- Compared product and category profitability.
- Identified products with high revenue but lower profit margins.

5. SQL Business Analysis

- Solved real-world business problems using SQL.
- Applied GROUP BY, HAVING, CASE WHEN, Subqueries, CTEs, and Window Functions to generate business insights.

6. Dashboard & KPI Analysis

- Built interactive dashboards in Excel and Power BI.
- Monitored KPIs including Revenue, Profit, Orders, Customers, Average Rating, and Monthly Sales Trends.

Business Insights

Key Business Insights

- Fashion was the highest revenue-generating product category.
- Pune generated the highest overall sales revenue.
- Returning customers contributed more revenue than new customers.
- Some categories generated high revenue but had relatively lower profit margins, indicating opportunities to improve profitability.

Recommendations

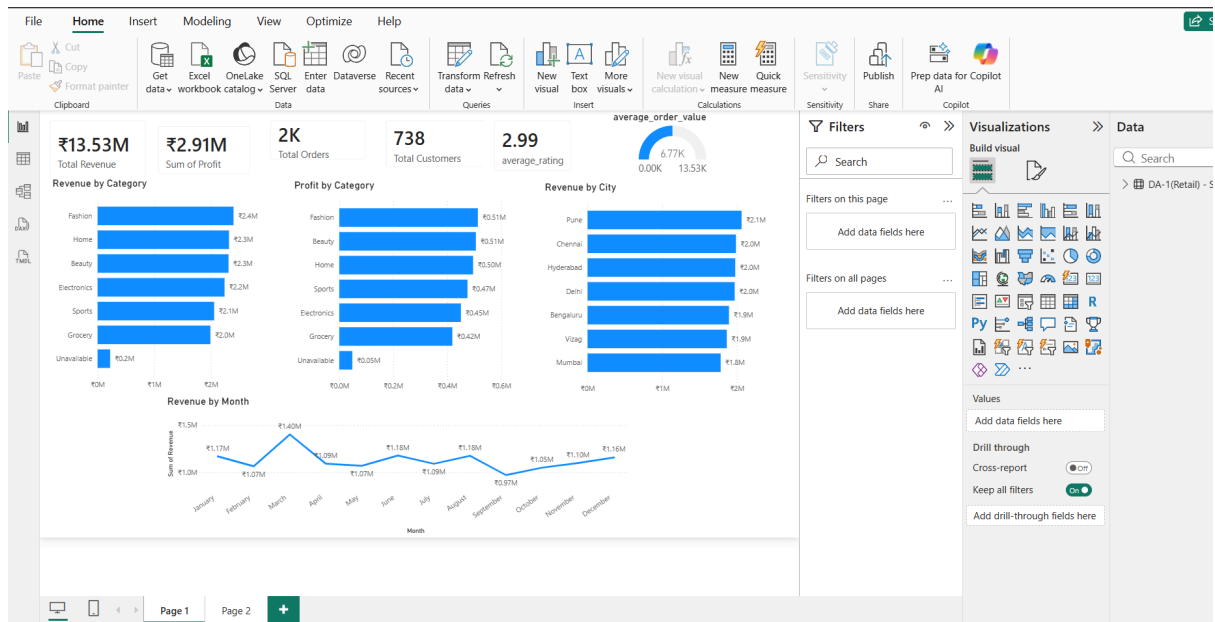
- Invest more in high-performing product categories.
 - Improve profitability in low-margin categories by reviewing pricing and costs.
 - Strengthen customer retention programs for returning customers.
 - Use monthly sales trends to improve inventory and marketing planning.
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Resume-Ready Project Summary

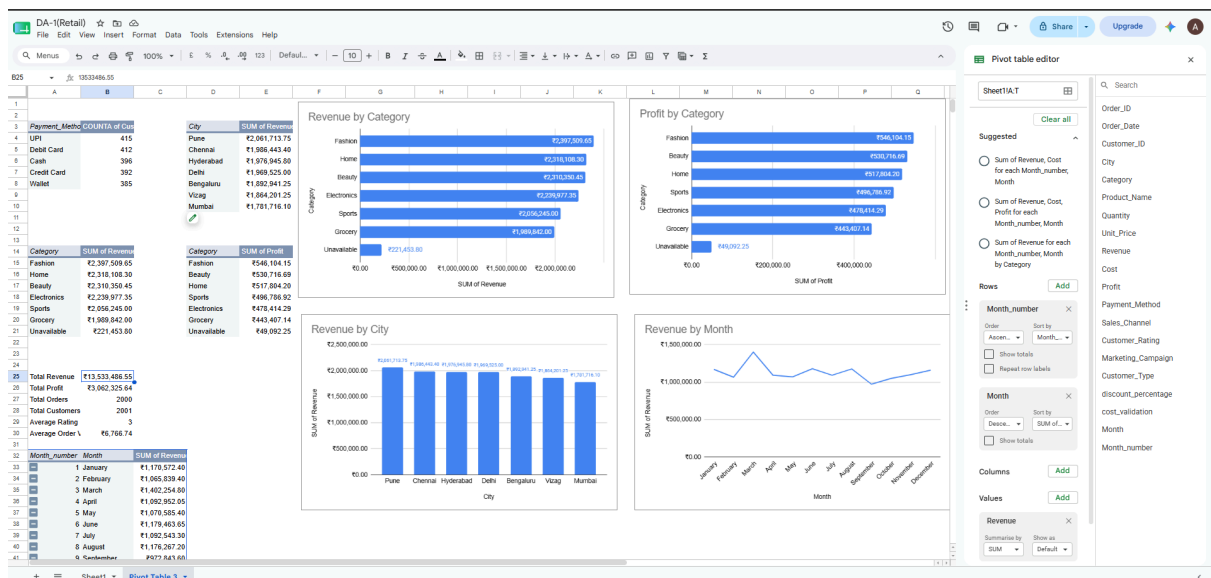
Retail Sales Analysis Project | SQL, Excel, Power BI

- Performed end-to-end analysis on retail sales data containing customer, product, sales, and transaction information.
 - Used SQL to solve real-world business problems using aggregate functions, GROUP BY, HAVING, subqueries, CTEs, and window functions.
 - Cleaned and prepared the dataset in Excel by handling missing values, validating data, and creating Pivot Tables for analysis.
 - Built an interactive Power BI dashboard to monitor revenue, profit, customer behavior, product performance, and monthly sales trends.
 - Developed KPI reports including Total Revenue, Total Profit, Total Orders, Total Customers, Average Rating, and Average Order Value.
 - Generated actionable business insights and recommendations to improve sales performance, customer retention, and product profitability.
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Power BI:



Sheets:



Conclusion

This project demonstrates an end-to-end retail sales analysis workflow using **Excel, SQL, and Power BI**. By combining data cleaning, business analysis, and interactive dashboards, the project provides actionable insights that can help businesses improve sales performance and support data-driven decision-making.

